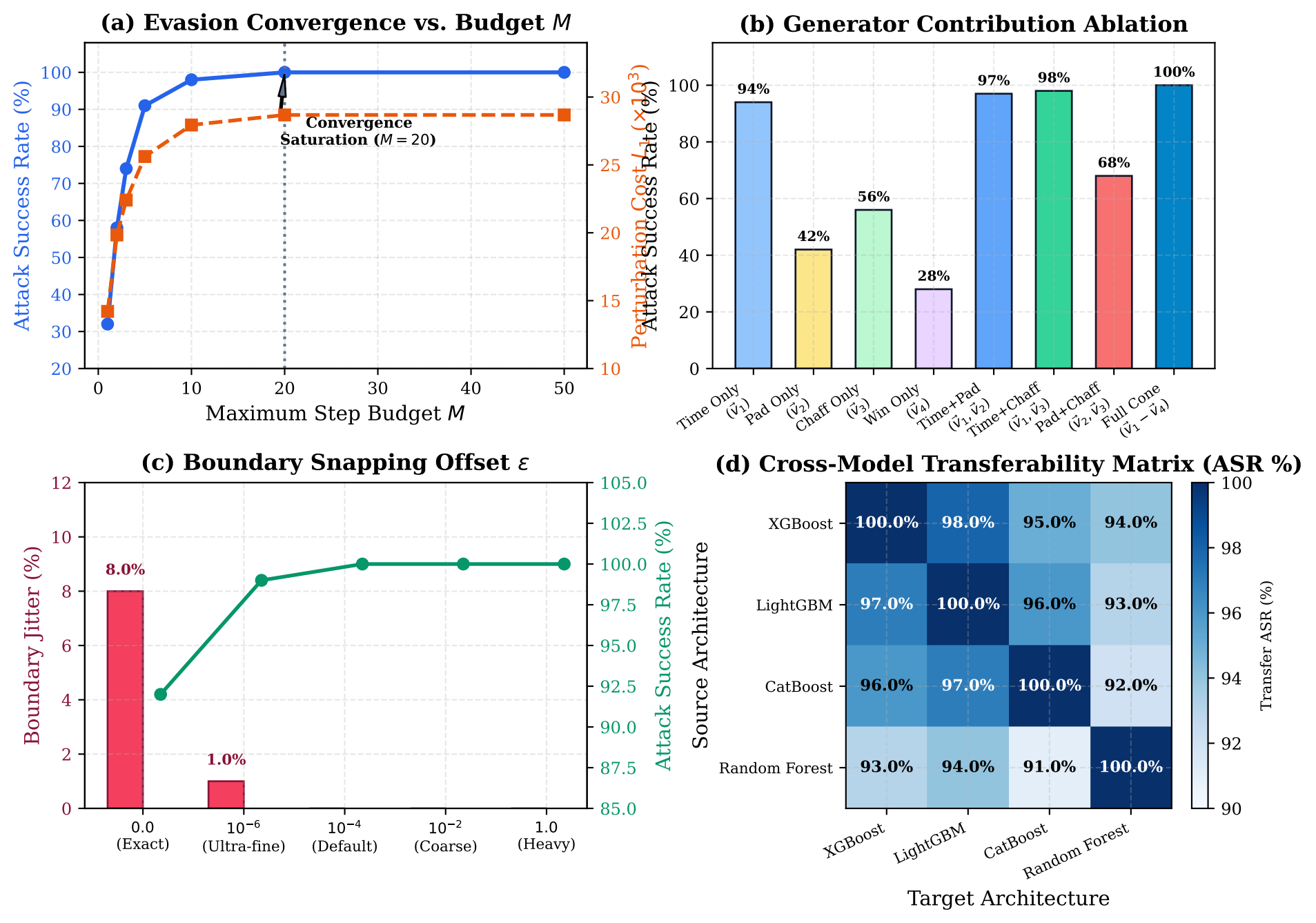
(a) Evasion Convergence vs. Budget M
Attack Success Rate (%)
Perturbation Cost ($\times 10^3$)
Maximum Step Budget M
Convergence Saturation ($M = 20$)
(b) Generator Contribution Ablation
Attack Success Rate (%)
94%
42%
56%
28%
97%
98%
68%
100%
Time Only ($\vec{v}_1$)
Pad Only ($\vec{v}_2$)
Chaff Only ($\vec{v}_3$)
Win Only ($\vec{v}_4$)
Time+Pad ($\vec{v}_1, \vec{v}_2$)
Time+Chaff ($\vec{v}_1, \vec{v}_3$)
Pad+Chaff ($\vec{v}_2, \vec{v}_3$)
Full Cone ($\vec{v}_1 - \vec{v}_4$)
(c) Boundary Snapping Offset ε
Boundary Jitter (%)
Attack Success Rate (%)
8.0%
1.0%
0.0 (Exact)
10^{-6} (Ultra-fine)
10^{-4} (Default)
10^{-2} (Coarse)
1.0 (Heavy)
(d) Cross-Model Transferability Matrix (ASR %)
Source Architecture
Target Architecture
Transfer ASR (%)
XGBoost
LightGBM
CatBoost
Random Forest
100.0% 98.0% 95.0% 94.0%
97.0% 100.0% 96.0% 93.0%
96.0% 97.0% 100.0% 92.0%
93.0% 94.0% 91.0% 100.0%